

Non homogeneous Second-Order ODEs

Recall when discussing Partial Fraction Decomposition in Calc II, we came up w/ terms like

$$\frac{1}{(x-1)^2(x+2)^3} = \frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{(x+2)} + \frac{D}{(x+2)^2} + \frac{E}{(x+2)^3}$$

So that when we multiplied everything out we would end up w/

$$1 = A(x-1)(x+2)^3 + B(x+2)^3 + C(x-1)^2(x+2)^2 + D(x-1)^2(x+2) + E(x-1)^2$$

Each term attached to the constants, the functions present are linearly independent. If they were not we would not have unique solutions for the constants.

The first method of solutions for nonhomogeneous ODEs of degree 2 (or higher) is called the Method of Undetermined Coefficients. It is essentially the same as "par-fac" but for ODEs.

This method only works if the forcing terms have finite number of linearly independent derivatives.

I.e. $m\ddot{x} + b\dot{x} + kx = f(t)$, where $f(t)$ is the forcing term.

Ex// $\cos(2x), -2\sin(2x), -4\cos(2x), \dots$ is not linearly independent, only the set containing $\{\cos(2x), -2\sin(2x)\}$ is linearly independent! Adding any more derivatives makes the collection dependent.

Ex// The linearly independent derivatives of x^3 are $3x^2$, $6x$, and 6 . Producing 0 automatically makes the collection dependent.

Ex// $\frac{1}{x}$ has an infinite number of linearly independent derivatives, and is thus not amenable to being solved using this method.

You're going to assume a solution of a specific form, plug it into the ODE, then match coeff. just like you would in a parfrac situation. That's why we need a finite number of linearly indep. vectors.

Ex// Find the general solution of $\ddot{x} + 4\dot{x} + 4x = 4t^2 + 6e^t$

The homogeneous eqn: $\ddot{x} + 4\dot{x} + 4x = 0$ produces a characteristic eqn $\lambda^2 + 4\lambda + 4 = 0$ or $(\lambda + 2)^2 = 0 \Rightarrow \lambda = -2$. This is a repeated real root, so this describes a critically damped motion. The solution always takes the form

$$y_h = (c_1 + c_2 t) e^{-2t}$$

We expect the particular solution to be a linear combination of the forcing terms and their linearly independent derivatives. So,

$$y_p = At^2 + Bt + C + De^t$$

Now take successive derivatives and plug into ODE to find conditions on A , B , C , and D to arrive at an identity.

$$\dot{y}_p = 2At + B + De^t$$

$$\ddot{y}_p = 2A + De^t$$

Therefore $\ddot{x} + 4\dot{x} + 4x = 4t^2 + 6e^t$

$$[2A + De^t] + 4[2At + B + De^t] + 4[At^2 + Bt + C + De^t] = 4t^2 + 6e^t$$

Since all terms are linearly independent, all coefficients on left must match those on the right.

Why? Justify this.

So, $[4A]t^2 + [4B + 8A]t + [4C + 4B + 2A]1 + [D + 4D + 4D]e^t = 4t^2 + 6e^t$

$$\Rightarrow 4A = 4$$

$$4B + 8A = 0 \quad \Rightarrow \quad A=1, B=-2, C=\frac{3}{2}, D=\frac{2}{3}$$

$$4C + 4B + 2A = 0$$

$$9D = 6$$

$$\therefore y_p = t^2 - 2t + \frac{3}{2} + \frac{2}{3}e^t$$

so that $y(t) = y_h + y_p = (c_1 + c_2 t)e^{-2t} + t^2 - 2t + \frac{3}{2} + \frac{2}{3}e^t$ ✓

Ex// Find the general solution of $\ddot{x} - 3\dot{x} + 2x = 2te^{3t} + 3\sin(t)$

The characteristic eqn. is $\lambda^2 - 3\lambda + 2 = 0 = (\lambda - 2)(\lambda - 1)$

$\Rightarrow \lambda = \{1, 2\}$, two real roots. Thus

$$y_h = c_1 e^t + c_2 e^{2t}$$

The forcing term is $f(t) = 2te^{3t} + 3\sin(t)$. The linearly independent derivatives include a combination of the following: $\{te^{3t}, e^{3t}, \sin(t), \cos(t)\}$. Thus, assume a solution of the form:

$$x_p = Ate^{3t} + Be^{3t} + C\sin(t) + D\cos(t)$$

$$\dot{x}_p = A(3t+1)e^{3t} + 3Be^{3t} + C\cos(t) - D\sin(t)$$

$$\ddot{x}_p = A[9t+6]e^{3t} + 9Be^{3t} - C\sin(t) - D\cos(t)$$

Plug this into the ODE to find conditions on the constants A, B, C, and D.

$$\ddot{x} - 3\dot{x} + 2x = 2te^{3t} + 3\sin(t)$$

$$[9Ate^{3t} + (6A+9B)e^{3t} - C\sin(t) - D\cos(t)]$$

$$-3[3Ate^{3t} + (A+3B)e^{3t} + C\cos(t) - D\sin(t)]$$

$$+ 2[Ate^{3t} + Be^{3t} + C\sin(t) + D\cos(t)] = 2te^{3t} + 3\sin(t)$$

Gather all like terms

$$[9A - 9A + 2A]te^{3t} + [6A + 9B - 3A - 9B + 2B]e^{3t} + [-C + 3D + 2C]\sin(t)$$

$$+ [-D - 3C + 2D]\cos(t) = 2te^{3t} + 3\sin(t)$$

Match Coeff.

$$2A = 2$$

$$3A + 2B = 0$$

$$A = 1, \quad B = -\frac{3}{2}, \quad C = \frac{3}{10}, \quad D = \frac{9}{10}$$

$$C + 3D = 3 \Rightarrow$$

$$D - 3C = 0$$

Thus,

$$y(t) = y_h + y_p = C_1e^t + C_2e^{2t} + te^{3t} - \frac{3}{2}e^{3t} + \frac{3}{10}\sin(t) + \frac{9}{10}\cos(t) \quad \checkmark$$

Ex // Find the general solution of $\ddot{x} - 3\dot{x} + 2x = 2t^2 + 3e^{2t}$

First the homogeneous solution:

The characteristic equation of the homogeneous ODE is

$$\begin{aligned}\lambda^2 - 3\lambda + 2 &= 0 \\ (\lambda - 2)(\lambda - 1) &= 0 \\ \Rightarrow \lambda &= \{1, 2\}\end{aligned}$$

$$x_h = C_1 e^t + C_2 e^{2t} \quad \checkmark$$

But notice, the forcing term already contains a term like e^{2t} ! So in coming up w/ a particular solution, since we have taken into account $t^0 e^{2t}$ through the homogeneous part, we consider $t^{0+1} e^{2t}$ and all unaccounted linearly independent derivatives. Thus,

$$\begin{aligned}x_p &= At^2 + Bt + C + Dte^{2t} \\ \dot{x}_p &= 2At + B + D(2t+1)e^{2t} = 2At + B + 2Dte^{2t} + De^{2t} \\ \ddot{x}_p &= 2A + 2D(2t+1)e^{2t} + 2De^{2t} = 2A + 4Dte^{2t} + 4De^{2t}\end{aligned}$$

Putting it all together:

$$\begin{aligned}[2A + 4Dte^{2t} + 4De^{2t}] - 3[2At + B + 2Dte^{2t} + De^{2t}] \\ + 2[At^2 + Bt + C + Dte^{2t}] = 2t^2 + 3e^{2t}\end{aligned}$$

$$\begin{aligned}\text{Gather like terms: } [2A]t^2 + [-6A + 2B]t + [2A - 3B + 2C]1 \\ + [4D - 6D + 2D]te^{2t} + [4D - 3D]e^{2t} = 2t^2 + 3e^{2t}\end{aligned}$$

Which implies

$$2A = 2$$

$$-6A + 2B = 0 \Rightarrow A = 1, B = 3, C = \frac{7}{2}, D = 3$$

$$2A - 3B + 2C = 0$$

$$D = 3$$

which means the general solution is:

$$x(t) = x_h + x_p = C_1 e^t + C_2 e^{2t} + t^2 + 3t + \frac{7}{2} + 3te^{3t}$$

Ex// Find the general solution of $\ddot{x} - 3\dot{x} + 2x = te^{2t} + \sin(t)$

Again, the homogeneous solution is $x_h = C_1 e^t + C_2 e^{2t}$

Notice the homogeneous solution has a term e^{2t} , and the forcing term contains te^{2t} . Thus we consider $t^n e^{2t}$ and all of its unaccounted linearly independent derivatives. Thus, we assume,

$$x_p = At^2 e^{2t} + Bte^{2t} + C \sin(t) + D \cos(t)$$

$$\begin{aligned} \dot{x}_p &= A(2t^2 + 2t)e^{2t} + B(2t+1)e^{2t} + C \cos(t) - D \sin(t) \quad \text{rewrite} \\ &= 2At^2 e^{2t} + (2A+2B)te^{2t} + Be^{2t} + C \cos(t) - D \sin(t) \end{aligned}$$

$$\begin{aligned} \ddot{x}_p &= 2A(2t^2 + 2t)e^{2t} + (2A+2B)(2t+1)e^{2t} + 2Be^{2t} - C \sin(t) - D \cos(t) \quad \text{rewrite} \\ &= 4At^2 e^{2t} + (8A+4B)te^{2t} + (2A+4B)e^{2t} - C \sin(t) - D \cos(t) \end{aligned}$$

Putting it all together,

$$\begin{aligned} &[4At^2 e^{2t} + (8A+4B)te^{2t} + (2A+4B)e^{2t} - C \sin(t) - D \cos(t)] \\ &- 3[2At^2 e^{2t} + (2A+2B)te^{2t} + Be^{2t} + C \cos(t) - D \sin(t)] \\ &+ 2[At^2 e^{2t} + Bte^{2t} + C \sin(t) + D \cos(t)] = te^{2t} + \sin(t) \end{aligned}$$

Gathering like terms:

$$2Ate^{2t} + (2A+B)e^{2t} + (C+3D)\sin(t) + (D-3C)\cos(t) = te^{2t} + \sin(x)$$

This implies:

$$\begin{aligned} 2A &= 1 & \Rightarrow A &= \frac{1}{2}, B = -1 \\ 2A+B &= 0 & C &= \frac{1}{10}, D = \frac{3}{10} \\ C+3D &= 1 \\ D-3C &= 0 \end{aligned}$$

$\therefore$ the general solution looks like

$$x(t) = x_h + x_p = C_1 e^t + C_2 e^{2t} + \frac{1}{2} t^2 e^{2t} - t e^{2t} + \frac{1}{10} \sin(t) + \frac{3}{10} \cos(t).$$

Ex// $\ddot{x} + x = \sin^3(t)$

The homogeneous solution is easy. It is: $x_h = C_1 \sin(t) + C_2 \cos(t)$.

The difficulty now is find what form to assume for y_p . Notice, $\sin^3(t)$ can be written much more easily!

$$\begin{aligned} \sin^3(t) &= \left(\frac{e^{it} - e^{-it}}{2i} \right)^3 \\ &= \frac{e^{3it} - 3e^{it} + 3e^{-it} - e^{-3it}}{-8i} \\ &= -\frac{1}{4} \left(\frac{e^{3it} - e^{-3it}}{2i} \right) + \frac{3}{4} \left(\frac{e^{it} - e^{-it}}{2i} \right) \\ &= -\frac{1}{4} \sin(3t) + \frac{3}{4} \sin(t) \quad \checkmark \end{aligned}$$

Since the homogeneous solution already contains a term like $\sin(t) = t^0 \sin(t)$, we must assume a particular solution that looks like:

$$x_p = A \sin(3t) + B \cos(3t) + Ct \sin(t) + Dt \cos(t)$$

cutting to the chase

$$\dot{x}_p = 3A \cos(3t) - 3B \sin(3t) + C t \cos(t) + (\sin(t) - Dt \sin(t) + D \cos(t))$$

$$\ddot{x}_p = -9A \sin(3t) - 9B \cos(3t) - C t \sin(t) + 2C \sin(t) - Dt \cos(t) - 2D \sin(t)$$

Putting it all together,

$$-8A \sin(3t) - 8B \cos(3t) + 2C \cos(t) - 2D \sin(t) = -\frac{1}{4} \sin(3t) + \frac{3}{4} \sin(t)$$

This gives: $-8A = -\frac{1}{4}$

$$-8B = 0$$

$$2C = 0$$

$$-2D = \frac{3}{4}$$

$$\Rightarrow A = \frac{1}{32} \quad B = C = 0 \quad D = -\frac{3}{8}$$

Therefore the general solution is given by:

$$X(t) = x_h + x_p = C_1 \sin(t) + C_2 \cos(t) + \frac{1}{32} \sin(3t) - \frac{3}{8} t \cos(t)$$

The Method of Variation of Parameters

This method is more versatile than the last one, in that it is applicable even if the forcing terms have an infinite number of linearly independent derivatives.

The method starts w/ a second-order linear ODE w/ constant coeff. (really, any order would work).

$$a_2 \ddot{x} + a_1 \dot{x} + a_0 x = f(t), \quad a_2 \neq 0$$

Here, our case is more general, we need not restrict ourselves to $a_2 = m$, $a_1 = b$, $a_0 = k$.

The primary assumption is that we can solve for the homogeneous case:

$$a_2 \ddot{x} + a_1 \dot{x} + a_0 x = 0,$$

and find two solutions: $x_h = C_1 x_1 + C_2 x_2$

The method of Variation of parameters works by literally varying the parameters!

We assume that: $C_1 \mapsto u_1(t)$ and $C_2 \mapsto u_2(t)$

We treat them as functions! Then we write

$$x_p = u_1(t) x_1(t) + u_2(t) x_2(t)$$

u_1 & u_2 are unknown functions we must solve for. To do this we take successive derivatives of x_p , plug them into the ODE, and get a condition for those functions.

Thus, $x_p = u_1 x_1 + u_2 x_2$

$$\dot{x}_p = (\dot{u}_1 \dot{x}_1 + \dot{u}_2 \dot{x}_2) + (\dot{u}_1 x_1 + \dot{u}_2 x_2)$$

$$\ddot{x}_p = (\ddot{u}_1 \ddot{x}_1 + \ddot{u}_2 \ddot{x}_2) + (\ddot{u}_1 \dot{x}_1 + \ddot{u}_2 \dot{x}_2) + \frac{d}{dt}(\dot{u}_1 x_1 + \dot{u}_2 x_2)$$

hold on to this!

Plug these into the original ODE: $a_2 \ddot{x} + a_1 \dot{x} + a_0 x = f(t)$

$$\begin{aligned} & a_2 (\ddot{u}_1 \ddot{x}_1 + \ddot{u}_2 \ddot{x}_2) + a_2 (\dot{u}_1 \dot{x}_1 + \dot{u}_2 \dot{x}_2) + a_2 \frac{d}{dt}(\dot{u}_1 x_1 + \dot{u}_2 x_2) \\ & + a_1 (\dot{u}_1 \dot{x}_1 + \dot{u}_2 \dot{x}_2) + a_1 (\dot{u}_1 x_1 + \dot{u}_2 x_2) \\ & + a_0 (u_1 x_1 + u_2 x_2) = f(t) \end{aligned}$$

Now we can cancel some terms! x_1 & x_2 solve the homogeneous ODE, so gather them up!

$$\begin{aligned} & u_1 (a_2 \ddot{x}_1 + a_1 \dot{x}_1 + a_0 x_1) + u_2 (a_2 \ddot{x}_2 + a_1 \dot{x}_2 + a_0 x_2) \\ & + a_2 (\dot{u}_1 \dot{x}_1 + \dot{u}_2 \dot{x}_2) + a_2 \frac{d}{dt}(\dot{u}_1 x_1 + \dot{u}_2 x_2) + a_1 (\dot{u}_1 x_1 + \dot{u}_2 x_2) = f(t) \end{aligned}$$

The magenta and green terms solve the homogeneous ODE, so they must be zero!

This next step is tricky. We have one equation with two unknowns: $\dot{u}_1$ & $\dot{u}_2$.

We can choose a condition (your book calls it the auxiliary condition) to simplify life.

$$\dot{u}_1 x_1 + \dot{u}_2 x_2 = 0 \quad !$$

Then we get $\dot{u}_1 x_1 + \dot{u}_2 x_2 = 0$

$$\dot{u}_1 \dot{x}_1 + \dot{u}_2 \dot{x}_2 = f(t)/a_2$$

This is a 2×2 system:

$$\begin{pmatrix} x_1 & x_2 \\ \dot{x}_1 & \dot{x}_2 \end{pmatrix} \begin{pmatrix} \dot{u}_1 \\ \dot{u}_2 \end{pmatrix} = \begin{pmatrix} 0 \\ f(t)/a_2 \end{pmatrix}$$

The cleanest way to solve this is to use Cramer's Rule. Thus:

$$\dot{u}_1 = \frac{\begin{vmatrix} 0 & x_2 \\ f(t)/a_2 & \dot{x}_2 \end{vmatrix}}{\begin{vmatrix} x_1 & x_2 \\ \dot{x}_1 & \dot{x}_2 \end{vmatrix}}$$

and

$$\dot{u}_2 = \frac{\begin{vmatrix} x_1 & 0 \\ \dot{x}_1 & f(t)/a_2 \end{vmatrix}}{\begin{vmatrix} x_1 & x_2 \\ \dot{x}_1 & \dot{x}_2 \end{vmatrix}}$$

Integrating both of these solves for $u_1(t)$ & $u_2(t)$, then we plug it back into our x_p to finish the problem:

$$x_p = u_1 x_1 + u_2 x_2$$

and

$$x(t) = x_h + x_p$$

Note the denominator of $\dot{u}_1$ & $\dot{u}_2$. It is the Wronskian. It is not zero because x_1 & x_2 are linearly independent by construction of the homogeneous solution!

We can rewrite

$$\dot{u}_1 = \frac{-x_2 f(t)/a_2}{W(x_1, x_2)} \quad \text{and} \quad \dot{u}_2 = \frac{x_1 f(t)/a_2}{W(x_1, x_2)}$$

Ex// Find the general solution of $\ddot{x} + 4\dot{x} + 4x = 3te^{-2t}$

The characteristic eqn is given by $\lambda^2 + 4\lambda + 4 = 0$

$$\text{so } (\lambda + 2)^2 = 0 \Rightarrow \lambda = -2$$

$$\therefore x_h = c_1 e^{-2t} + c_2 t e^{-2t} \quad \text{w/ } x_1 = e^{-2t}, \quad x_2 = t e^{-2t}$$

We assume that $x_p = u_1 e^{-2t} + u_2 t e^{-2t}$, and $a_2 = 1$, $f(t) = 3te^{-2t}$

$$\text{Therefore } \dot{u}_1 = \frac{-x_2 f(t)/a_2}{W} \quad \text{and} \quad \dot{u}_2 = \frac{x_1 f(t)/a_2}{W}$$

$$W = \begin{vmatrix} e^{-2t} & t e^{-2t} \\ -2e^{-2t} & (-2t+1)e^{-2t} \end{vmatrix} = e^{-4t} \begin{vmatrix} 1 & t \\ -2 & (-2t+1) \end{vmatrix} = e^{-4t} ((-2t+1) + 2t) = e^{-4t}$$

$$\therefore \dot{u}_1 = \frac{-t e^{-2t} (3t e^{-2t})}{e^{-4t}} = -3t^2 \Rightarrow u_1 = -t^3$$

$$\dot{u}_2 = \frac{e^{-2t} (3t e^{-2t})}{e^{-4t}} = 3t \Rightarrow u_2 = \frac{3}{2} t^2$$

$$\begin{aligned} \therefore x(t) = x_h + x_p &= \left[c_1 e^{-2t} + c_2 t e^{-2t} \right] + \left[\overset{u_1}{\underset{\downarrow}{-t^3}} \left(\overset{x_1}{\underset{\downarrow}{e^{-2t}}} \right) + \overset{u_2}{\underset{\downarrow}{\frac{3}{2} t^2}} \left(\overset{x_2}{\underset{\downarrow}{t e^{-2t}}} \right) \right] \\ &= \left[c_1 e^{-2t} + c_2 t e^{-2t} \right] + \left[\frac{1}{2} t^3 e^{-2t} \right] ! \end{aligned}$$

Ex// Find the general solution of $\ddot{x} + x = \tan(t)$, $t \in (-\frac{\pi}{2}, \frac{\pi}{2})$

The homogeneous solution is easy: $x_h = C_1 \cos(t) + C_2 \sin(t)$.

Assume the following form: $x_p = u_1 \cos(t) + u_2 \sin(t)$.

$$x_1 = \cos(t), \quad x_2 = \sin(t), \quad f(t) = \tan(t), \quad a_2 = 1$$

$$W = \begin{vmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{vmatrix} = 1$$

$$\dot{u}_1 = \frac{-x_2 f(t)/a_2}{W} \quad \text{and} \quad \dot{u}_2 = \frac{x_1 f(t)/a_2}{W}$$

$$\dot{u}_1 = -\sin(t) \tan(t)$$

$$\dot{u}_2 = \cos(t) \tan(t)$$

$$u_1 = -\int \frac{\sin^2(t)}{\cos(t)} dt = \int \frac{\cos^2(t) - 1}{\cos(t)} dt = \int \cos(t) - \sec(t) dt = \sin(t) - \ln|\sec(t) + \tan(t)|$$

$$u_2 = \int \sin(t) dt = -\cos(t).$$

$$\text{Thus: } x(t) = \underbrace{C_1 \cos(t) + C_2 \sin(t)}_{x_h} + \overbrace{\left(\sin(t) - \ln|\sec(t) + \tan(t)| \right) \cos(t)}^{u_1} + \underbrace{(-\cos(t)) \sin(t)}_{\substack{\uparrow u_2 \quad \uparrow x_2}}$$

$$x(t) = C_1 \cos(t) + C_2 \sin(t) - \cos(t) \ln|\sec(t) + \tan(t)| \quad \checkmark$$